

Krishna Defence / Garuda v3

SIH26153 - Network traffic world model

Architecture and validation boundary - 11 September 2026

1. Observed telemetry and graph construction

CIC CSV records enter the network state at **flow end time**, using timestamp plus duration. This avoids treating completed-flow statistics as available when the connection started. The supplied captures produce 6,453 closed-window service graphs. Repeated headers are ignored, invalid numeric values fail ingestion, and unknown labels remain unknown.

A service graph links observed protocols to destination-service categories. It does not reconstruct missing hosts. Endpoint-preserving CSVs and classic Ethernet/raw-IPv4 PCAPs can form host graphs. The packet parser extracts TTL statistics, TCP window values, fragment indicators and duplicate payload segments. PCAP graph forecasting requires a checkpoint trained on packet telemetry.

2. Learned state transition model

Stage	Implementation
Spatial encoder	Two directed GraphSAGE layers with incoming and outgoing mean aggregation, masked node features and graph pooling.
Temporal encoder	LSTM over eight observed 10-second graph windows.
Autoregressive decoder	Four future windows. Each step outputs a diagonal Gaussian state distribution and a malicious-flow probability.
Learning	Reverse-mode differentiation through graph layers and LSTM. Joint future-state likelihood and horizon-specific binary loss.

The reference model uses NumPy with explicit reverse-mode automatic differentiation. Numerical finite-difference tests validate gradients through graph and recurrent weights. The decoder predicts pooled feature-state dynamics, not future host topology. Node identities and future labels are never model inputs.

3. Serving and explainability

A local authenticated HTTP service loads a versioned, hash-checked checkpoint. Each request carries a complete, ordered observation history, eliminating shared inference buffers. The offline dashboard displays the graph, forecast timeline and gradient-times-input feature sensitivity. It explicitly reports unknown attack stage because Bot/Benign labels do not supervise MITRE progression.

Evidence and response controls

4. Reproducible evaluation

Raw time windows split chronologically into train, validation and test before sequence construction. No context or target crosses a split or capture boundary. Missing windows break sequences. The primary run uses seed 42 and 456 / 158 / 160 examples. Validation loss selects checkpoints; validation F1 selects thresholds. Extra seeds 7 and 19 assess initialization sensitivity, without selecting the best test seed.

Model	F1	Precision	Recall	FPR
Logistic regression	91.6%	98.1%	86.0%	5.1%
LSTM	93.4%	99.1%	88.4%	2.6%
GNN + LSTM	94.4%	98.2%	90.9%	5.1%

Interpretation: These scores concern malicious traffic in the fourth future window, 40 seconds after the observation cutoff. They do not measure compromise prediction. The clean-history test contains one future positive; every model misses it. GNN future-state MSE is 0.05371 versus 0.01274 for persistence. Improving dynamics and collecting incident timelines remain necessary.

Across three seeds, GNN + LSTM mean F1 is 96.2% with 1.25 percentage-point population standard deviation. Overlapping windows are correlated, and a single dataset family limits generalization. Local checkpoint hashes and source-data hashes establish reproducibility, not independent provenance certification.

5. Bounded containment

Viewer and operator bearer tokens are separate. Host/origin checks, body limits and a bounded inference slot protect the local service. Response defaults to dry-run. Operators can publish only explicitly allowed IP scopes, with 10-900 second expiry. The Node proxy checks HMAC-signed policies. Revoke and kill-switch actions remove v3 policies. A SQLite hash chain records changes.

Tests verify real local selected-IP blocking, an unaffected comparison IP, restoration after revocation and rejection of a forged policy signature. Invalid policies fail open for this optional overlay; the existing WAF remains active. This tradeoff prevents a broken policy file from causing blanket denial, but requires operational alerting in production.

6. Deployment boundary and next evidence

This remains a single-instance research prototype. Enterprise deployment needs TLS termination, protected secrets, centralized identity/RBAC, independent penetration review, resource/load testing and protected audit anchoring. Host/packet-trained forecasting, supervised MITRE stages, unseen-campaign evaluation and incident-onset recall are open gates. No automatic action follows an unvalidated forecast.

Sources: project source and [garuda_v3/artifacts/run/metrics.json](#); official problem statement: <https://www.sih.gov.in/sih2026PS>; GraphSAGE operator: https://pytorch-geometric.readthedocs.io/en/latest/generated/torch_geometric.nn.conv.SAGEConv.html